Development of a Neural Network Framework for Learning Constitutive Laws in 2D Granular Flow

Name: Abhishek Tagalpallewar || 2nd Year Undergraduate || Course Code: ME 299 || Institute: IIT Gandhinagar || Supervisor: Harkirat Singh || Poster ID: 93

Introduction & Problem Statement

Objective: Establish neural network (NN) framework capable of learning the constitutive laws governing granular flow.

Methodology: By approximating fundamental mathematical functions before transitioning to complex experimental and simulation data, we evaluate how different architectures capture underlying physical relationships.

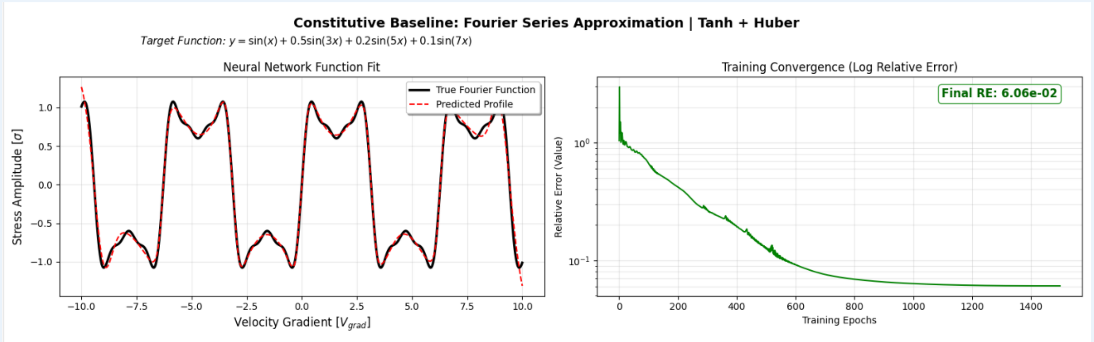
What is a Neural Network?

A neural network is a computational model that learns patterns in data by adjusting weights and biases applied to inputs through interconnected layers of neurons. During training, the predicted output is compared with the true value using a loss function, and the weights and biases are updated through backpropagation to minimize the prediction error

Loss = $(1/n) \sum |\text{True Value} - \text{Predicted Value}|$

Phase I: Basic Function Approximation

Testing Complexity: Shallow and deep networks were benchmarked on fundamental nonlinear functions to assess learning limits.
Quadratic $y=(x+2)^2$: Evaluated the capture of polynomial curvature.
Sine $y=\sin(3x)$: Evaluated performance on periodic and oscillatory patterns
Advanced: Exponential, Logarithmic, Fourier series
The Challenge: Fourier series involve high-frequency oscillations and multiple sharp peaks, testing the model's ability to learn complex, non-monotonic periodic patterns
Finding: A 3-layer architecture provided the optimal balance between modeling capacity and computational complexity.



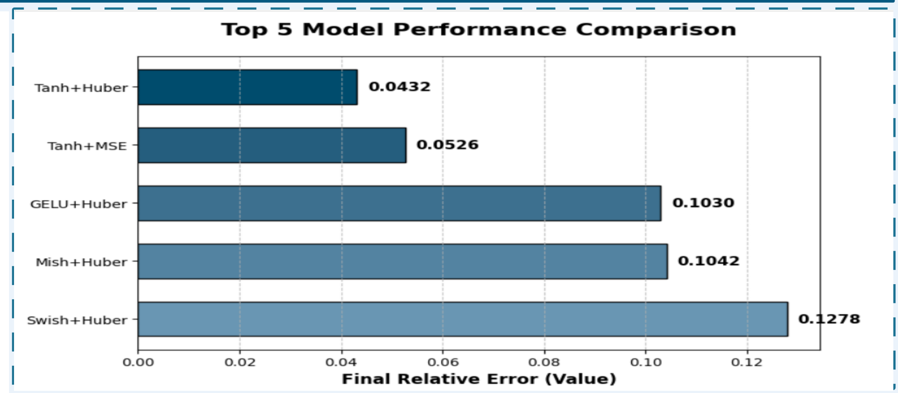
Phase II: Architectural Optimisation (Tournament)

A systematic Tournament identified the optimal model combination:

- **25 Combinations:** 5 Activations (ReLU, Tanh, Swish, Mish, GELU) x 5 Losses (MSE, MAE, Huber, MSLE, RE)
- **Key Finding:** 3-layer arch (64 nodes) optimal; Log-RE tracks convergence better than MSE

🏆 --- TOP 5 RANKED FOR FOURIER --- 🏆

Act	Loss	Final_RE	Time_Sec
Tanh	Huber	0.043155	0.53
Tanh	MSE	0.052591	0.56
GELU	Huber	0.103007	0.64
Mish	Huber	0.104158	1.48
Swish	Huber	0.127837	0.59

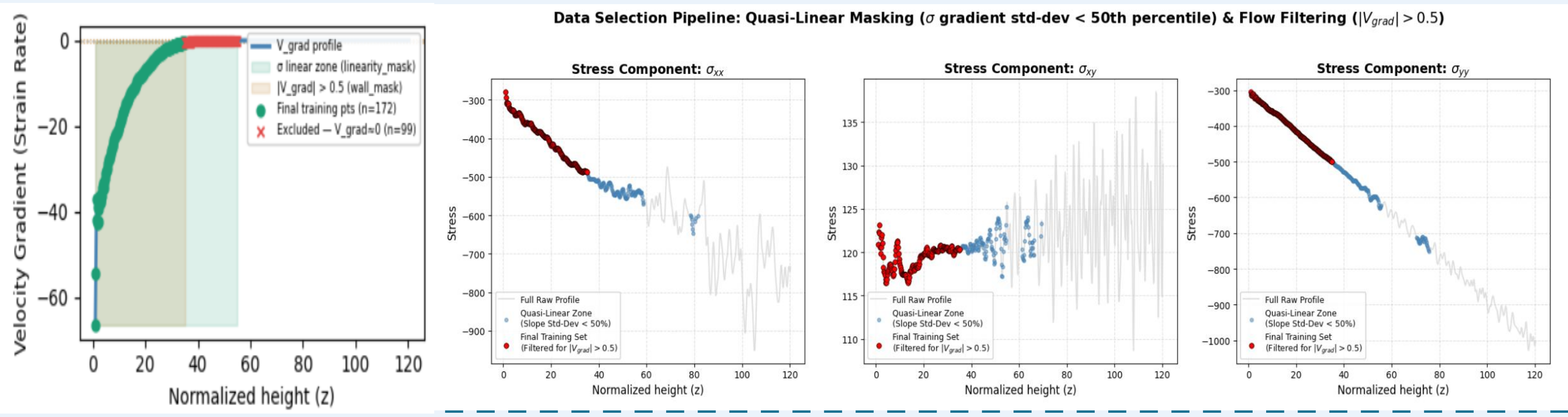


Phase III: Granular Data Pipeline

Constitutive Mapping : The model learns mapping from Velocity gradient (strain field) to stress Tensor using experimental data.

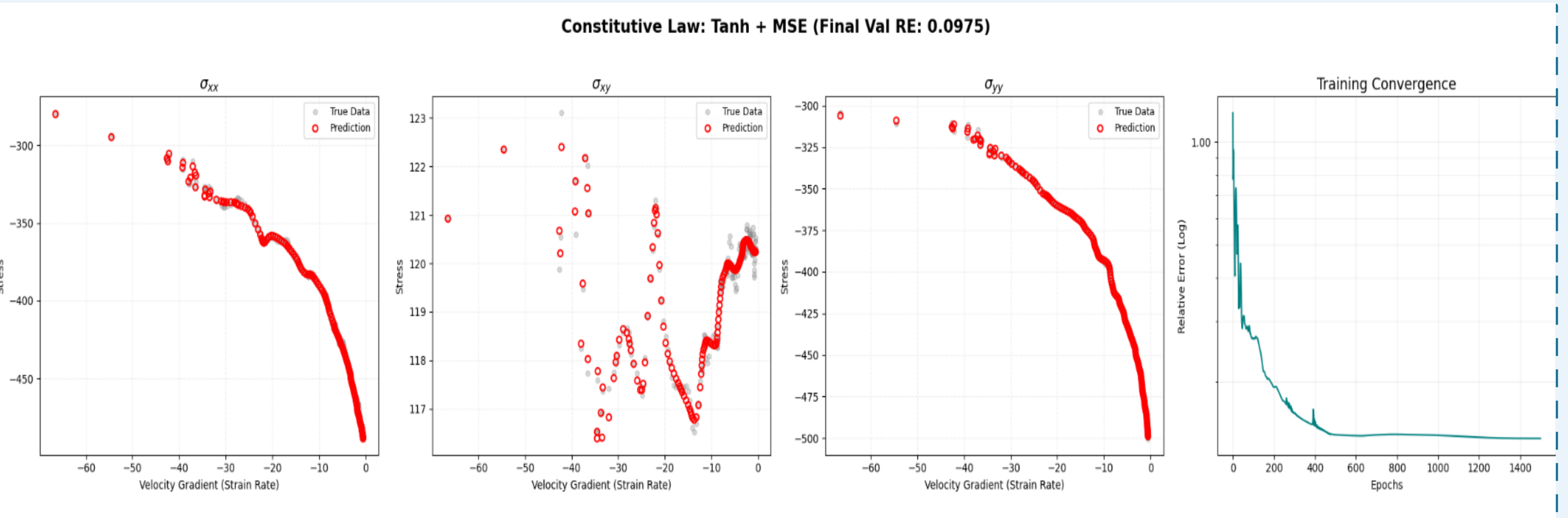
Quasi-Linear Masking: A rolling standard deviation mask identifies stable flow regions, filtering out high-noise segments and wall-boundary interference.

Data Augmentation: Gaussian noise injection was used to double training points, enhancing model robustness against experimental irregularities.



Phase IV : Constitutive Law - Training Results

- Constitutive Mapping : The model maps velocity gradients to the full stress tensor.
- Performance : Achieved high-fidelity interpolation with a relative error of around 10%.



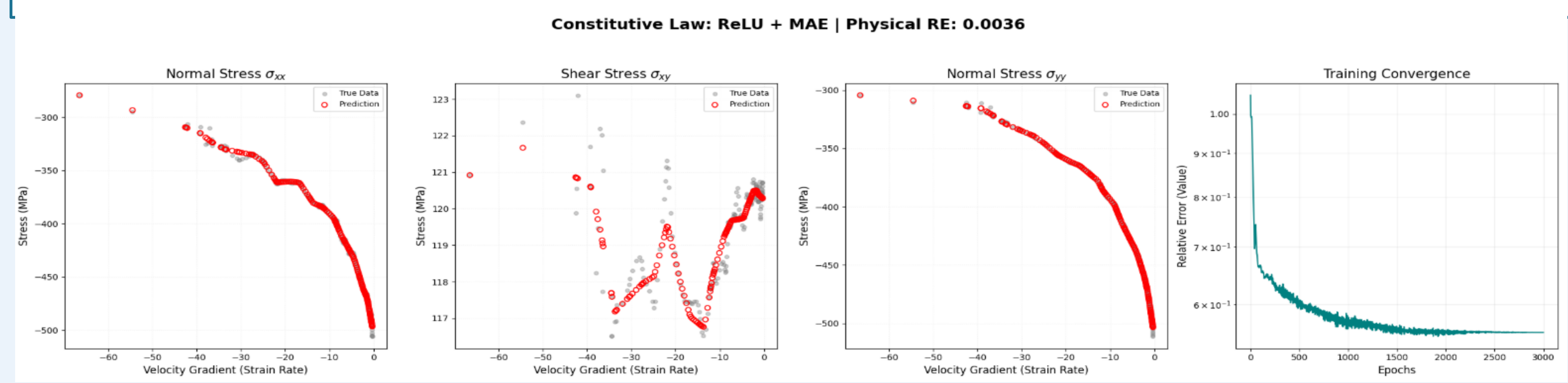
Gravity Present

Phase V: Transfer Learning & Validation

Following evaluation, the top-performing model was selected, and its parameters were fixed to function as a stable computational model for physical laws.

Validation: Tested on granular_data_3.(A new data set) Initially, direct deployment yielded a 48% Relative Error due to scale differences.

The Breakthrough: Implementing Reference Normalization (scaling by the mean of the new dataset) reduced the error to 3-4 %



Validation Set 1 :Gravity Present

Key Observations

Optimal Depth: A 3-layer architecture provides the necessary model capacity to resolve stress fluctuations. Shallow networks (1-2 layers) fail to capture the physics (underfitting), while deeper networks (>4 layers) provide no significant gain.

Scale Invariance: Direct model deployment across different datasets is scale-dependent. Implementing Reference Normalization effectively decouples the learned physical law from absolute values, reducing cross-dataset error from 48% to 3%

Domain Validity: Accuracy is high during interpolation (within the training range) but degrades during extrapolation.

Activation Performance: While ReLU is suitable for linear trends, Tanh and Mish activations are superior for mapping granular constitutive laws. Their smooth gradients allow the network to learn complex, non-linear stress patterns more effectively.

References

→Constitutive Mapping: Raissi et al., J. Comput. Phys. (2019).
→Activations (Mish/Tanh): Misra, D., arXiv:1908.08681 (2019).
→Optimization (Cosine Annealing): Loshchilov & Hutter, ICLR (2017).
→Normalization & Scaling: Ioffe & Szegedy, ICML (2015).
→Granular Physics: Saitoh & Mizuno, Phys. Rev. E (2016)

Key Result: Normalization

Cross-dataset Relative Error: 48% to 3%

Future Work

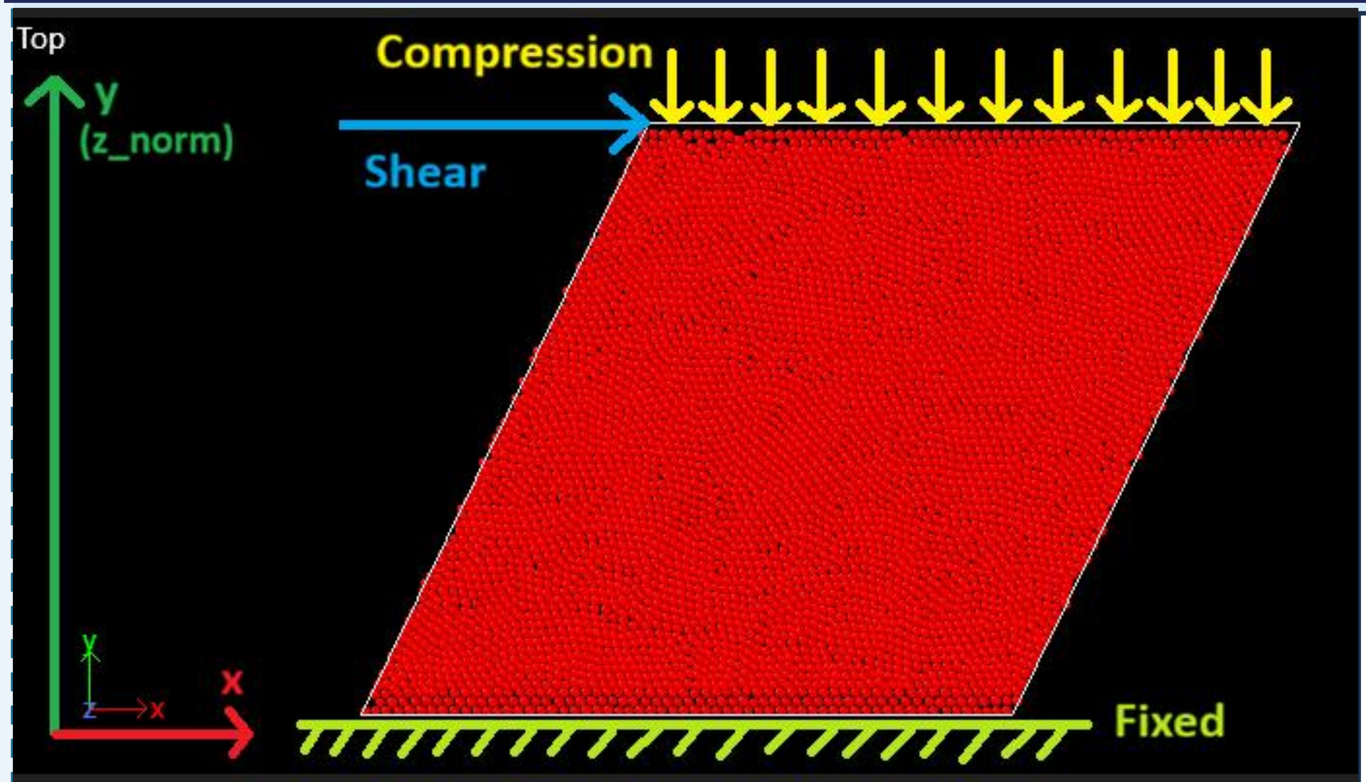
Building a Model Library: Create different models for different situations, such as very dense flows or very fast flows.

Improving Bulk Accuracy: The next step is to train the neural network directly on "Bulk" simulation data to get the error below 5%

Industrial Testing: Compare how these models perform in real-world tasks, like moving grain in a hopper or sand on a conveyor belt .

Fast Predictions: Because these models are "frozen" and fast, they can be used to predict stresses in real-time, saving the hours of time usually needed for a full simulation.

Performance Analysis & Physical Interpretation



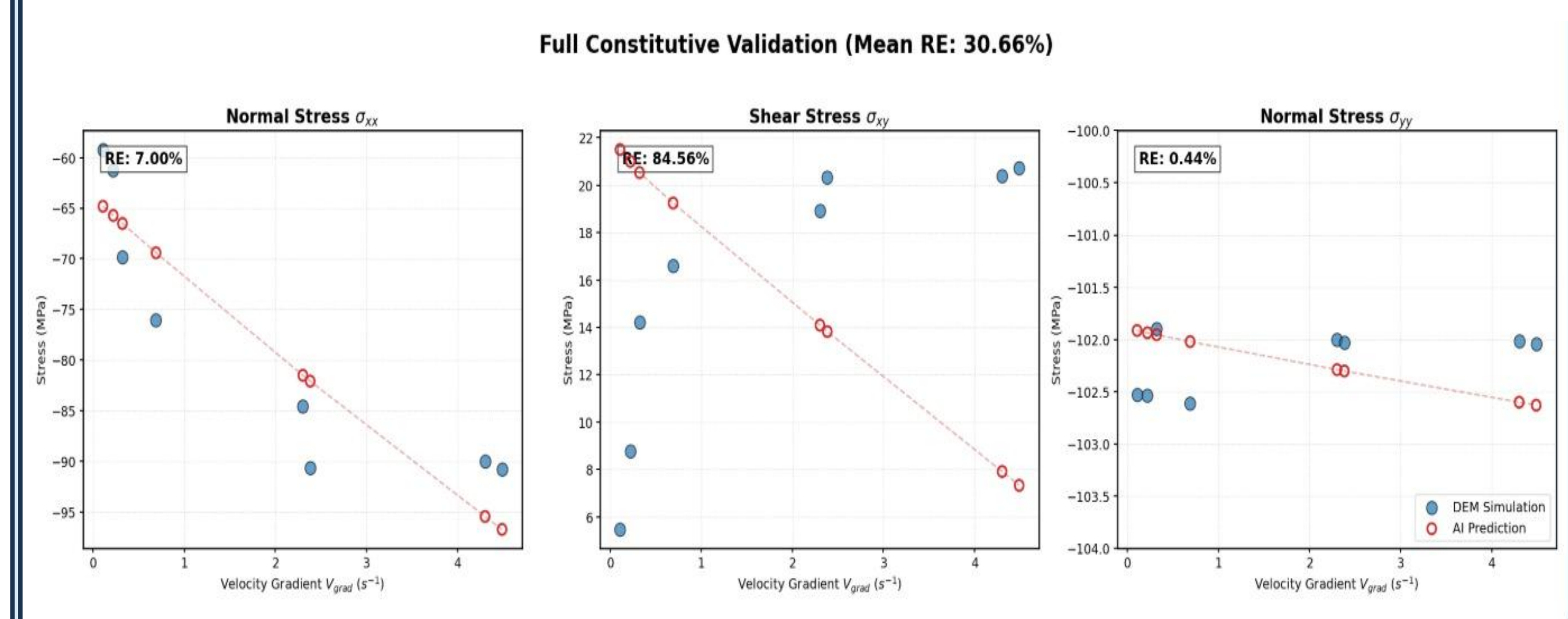
LAMMPS Simulation : 2D Granular Flow (Poster ID : 95)

Same Friction, Different Pressure: The friction coefficient is constant, but the source of pressure differs. In training, pressure comes from gravity (weight of particles), while in validation(Set 2) it arises only from boundary conditions.

Model Bias: Since the model uses only the velocity gradient as input, it learns a depth-dependent stress relationship from gravity-based training. This assumption breaks down in zero-gravity, where pressure is not linked to depth.

Magnitude vs. Trend: The model captures the trend of stress variation but fails to predict the correct magnitude (84.56% error) due to the absence of pressure as an input.

Normal Stress: Normal stress remains accurate (<10% error) because it reflects local inter-particle confinement, which is still reasonably inferred from the velocity gradient.



Validation Set 2 : Gravity Not Present